Kinvecta

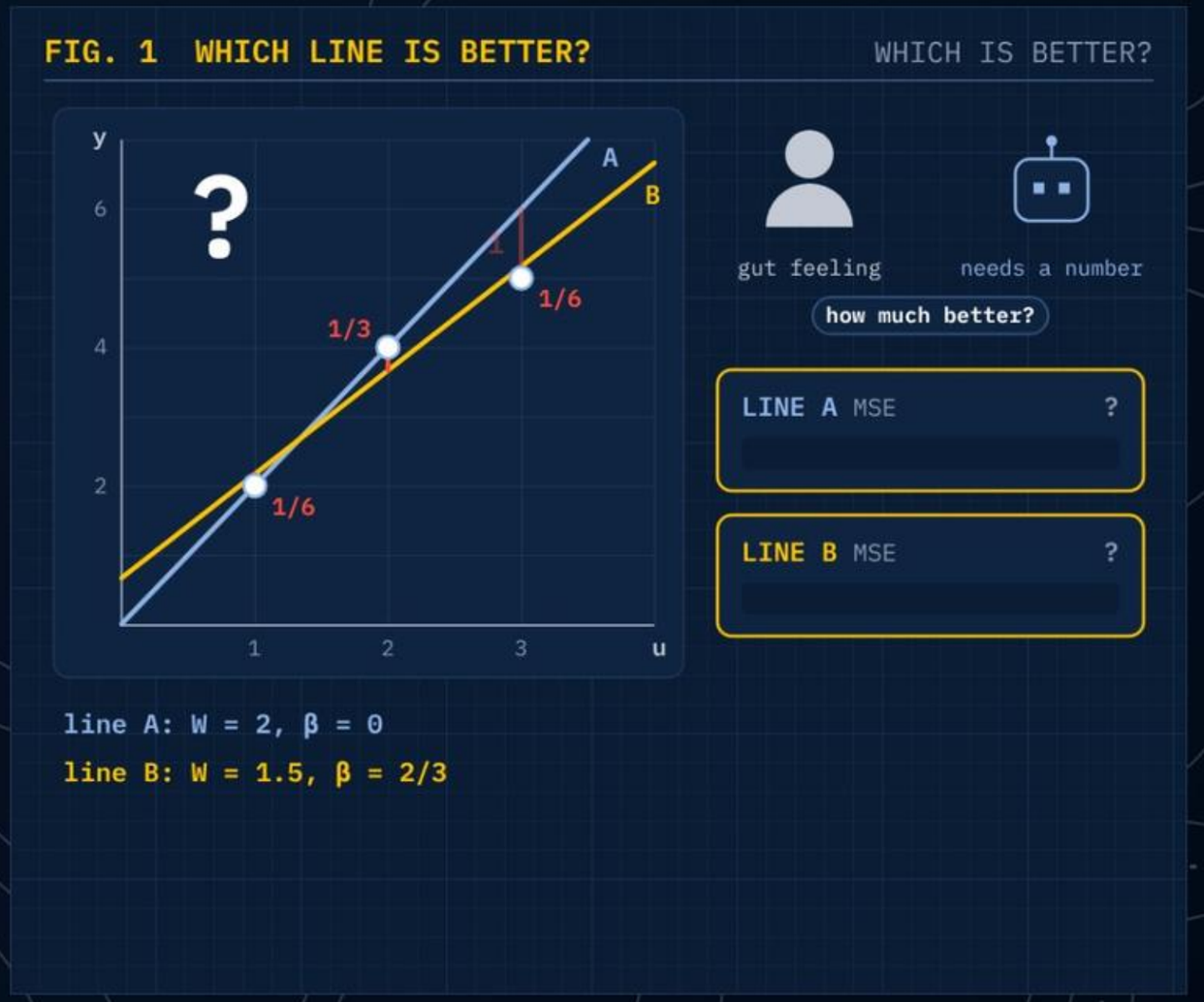
HOW LARGE LANGUAGE MODELS WORK · EPISODE 7 ·

Two lines. Which is better?

A computer can't go on a gut feeling. It needs a number, and how much better. Today we work it out by hand.

11-minute lesson

Companion notebook



MARK

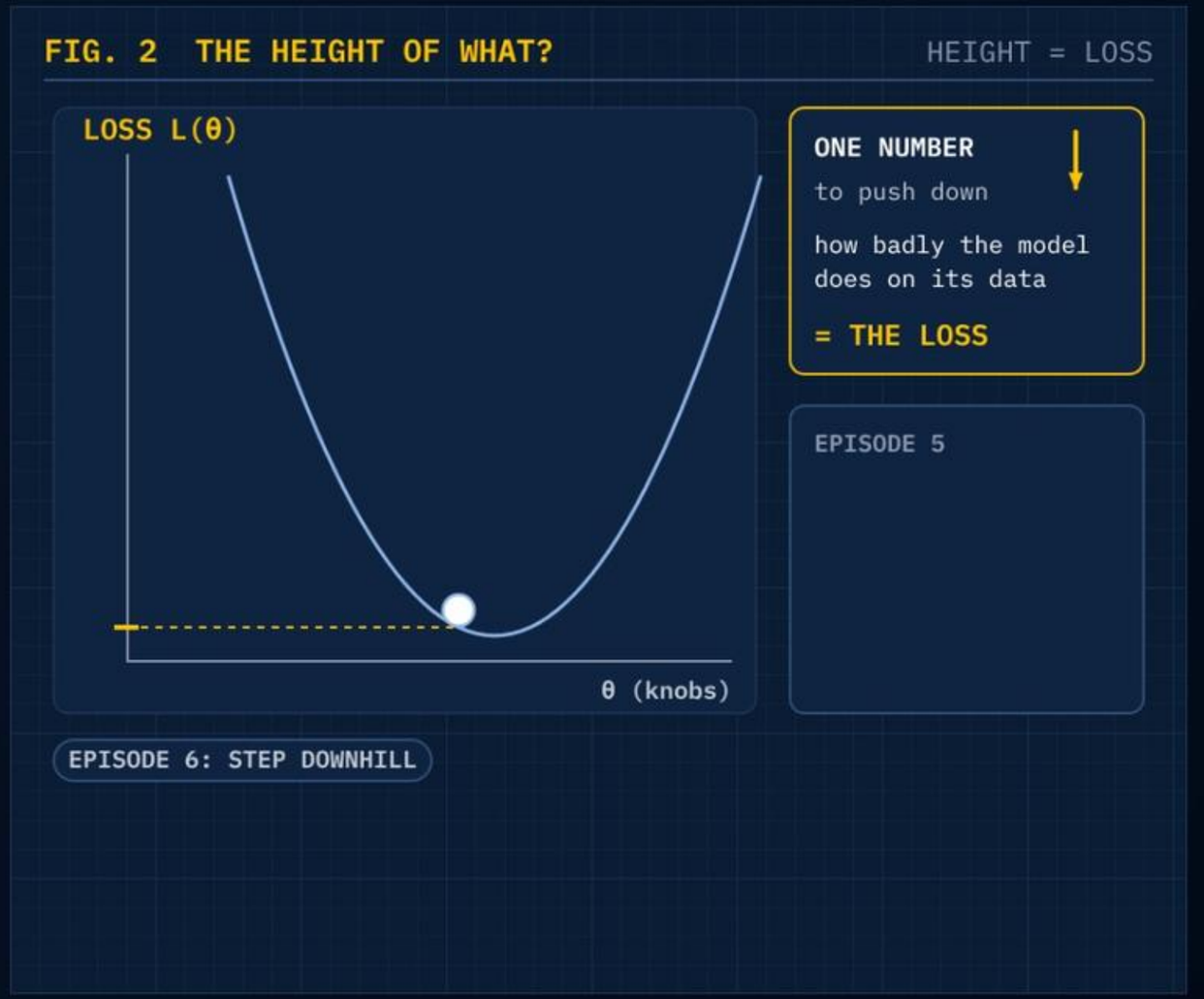
WHY THIS MATTERS

The height of what?

Episode 6 stepped downhill. Training needs one number to push down: the loss.

01 A gradient step lowers a height

02 For training, that height is the loss



Episodes 4–6. Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

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EPISODE 7 · WHAT LEARNING MEANS, PART 1

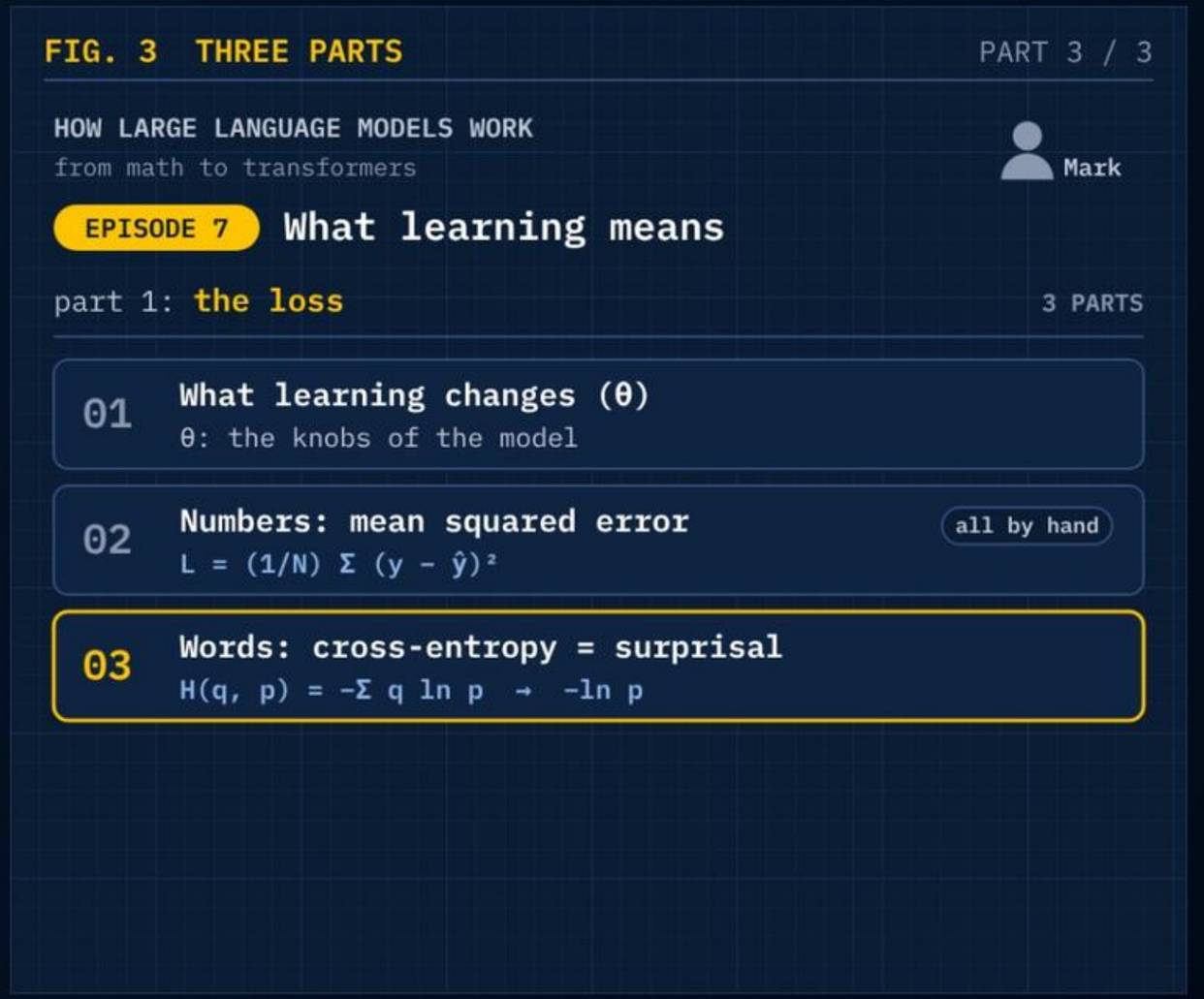
The loss, by hand

Three parts, every number worked by hand, and a notebook with all of them.

01 What learning changes

02 Numbers: mean squared error

03 Words: cross-entropy, which is surprisal



Companion notebook (Colab), linked in the description. Sources on every slide.

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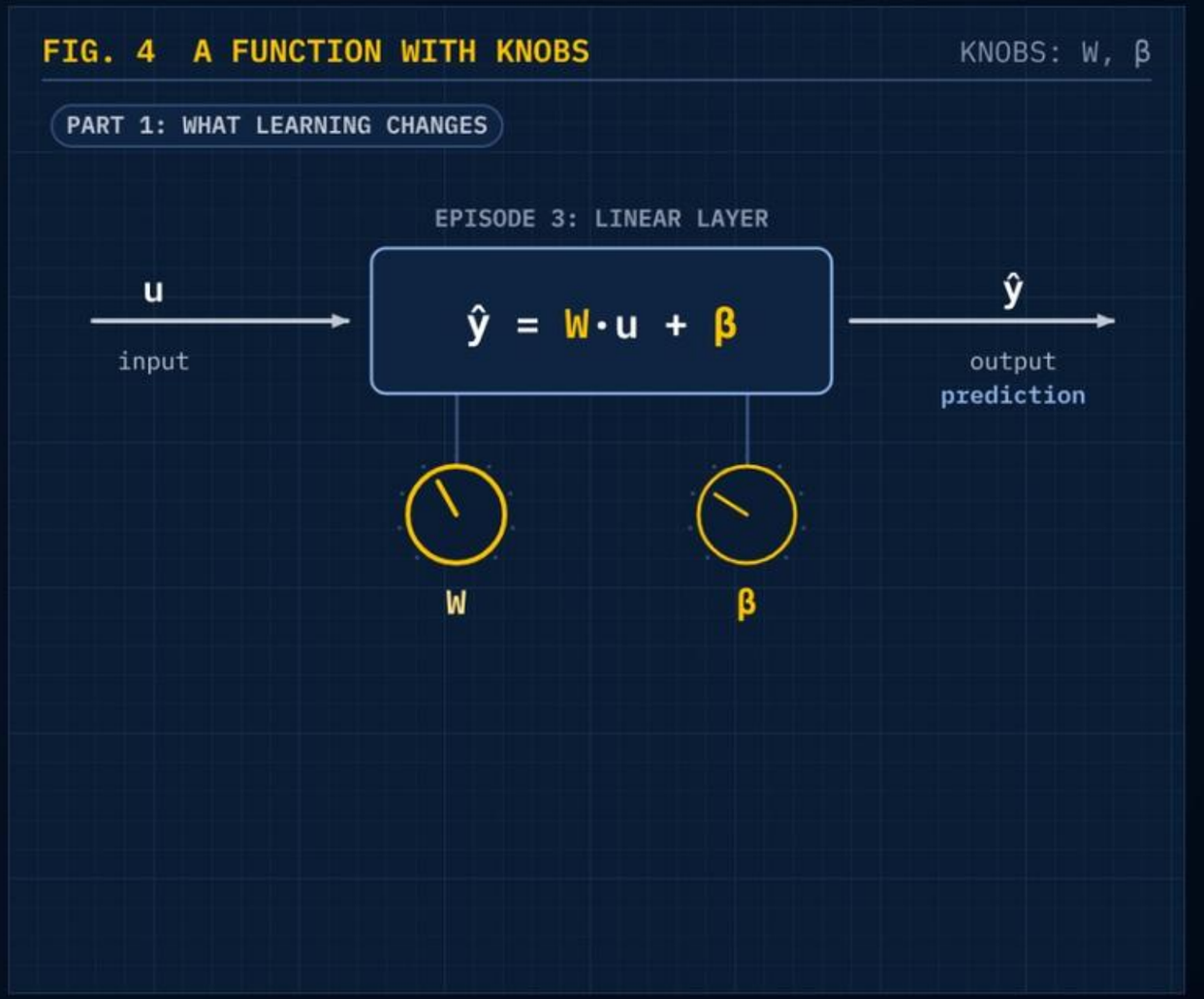
PART 1 · WHAT LEARNING CHANGES

A model is a function with knobs

$\hat{y} = W \cdot u + \beta$ (Episode 3's linear layer)

01 u: the input; ŷ: the prediction

02 W and β: the knobs



Episode 3 (linear layer). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 (maximum likelihood), eq. 5.66 (mean squared error).

PART 1 · TWO WORDS

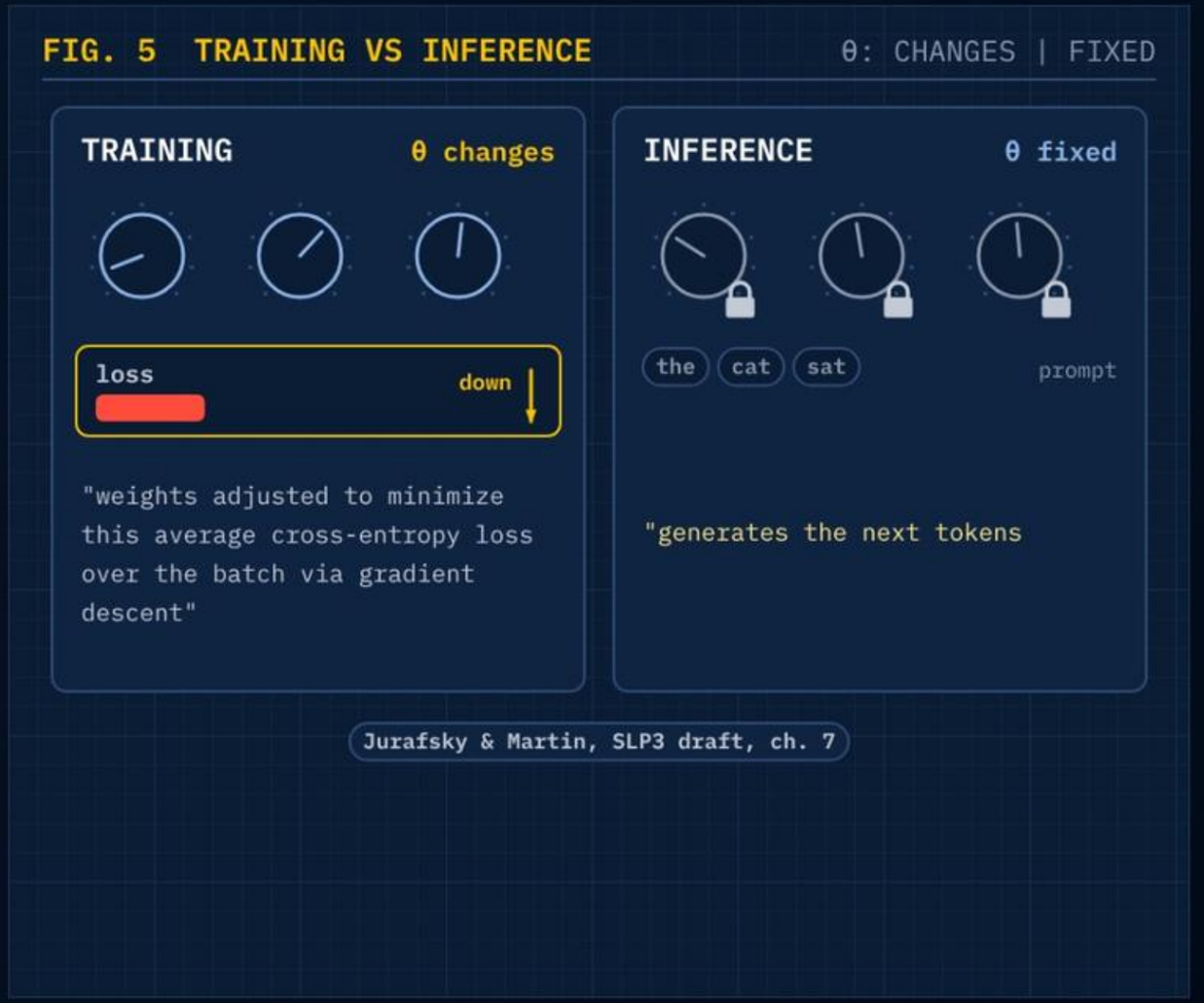
Training vs inference

Learning means turning the knobs to make the loss small.

01 Training changes θ

02 Inference runs the model with θ fixed

03 Weights “adjusted to minimize this average cross-entropy loss”



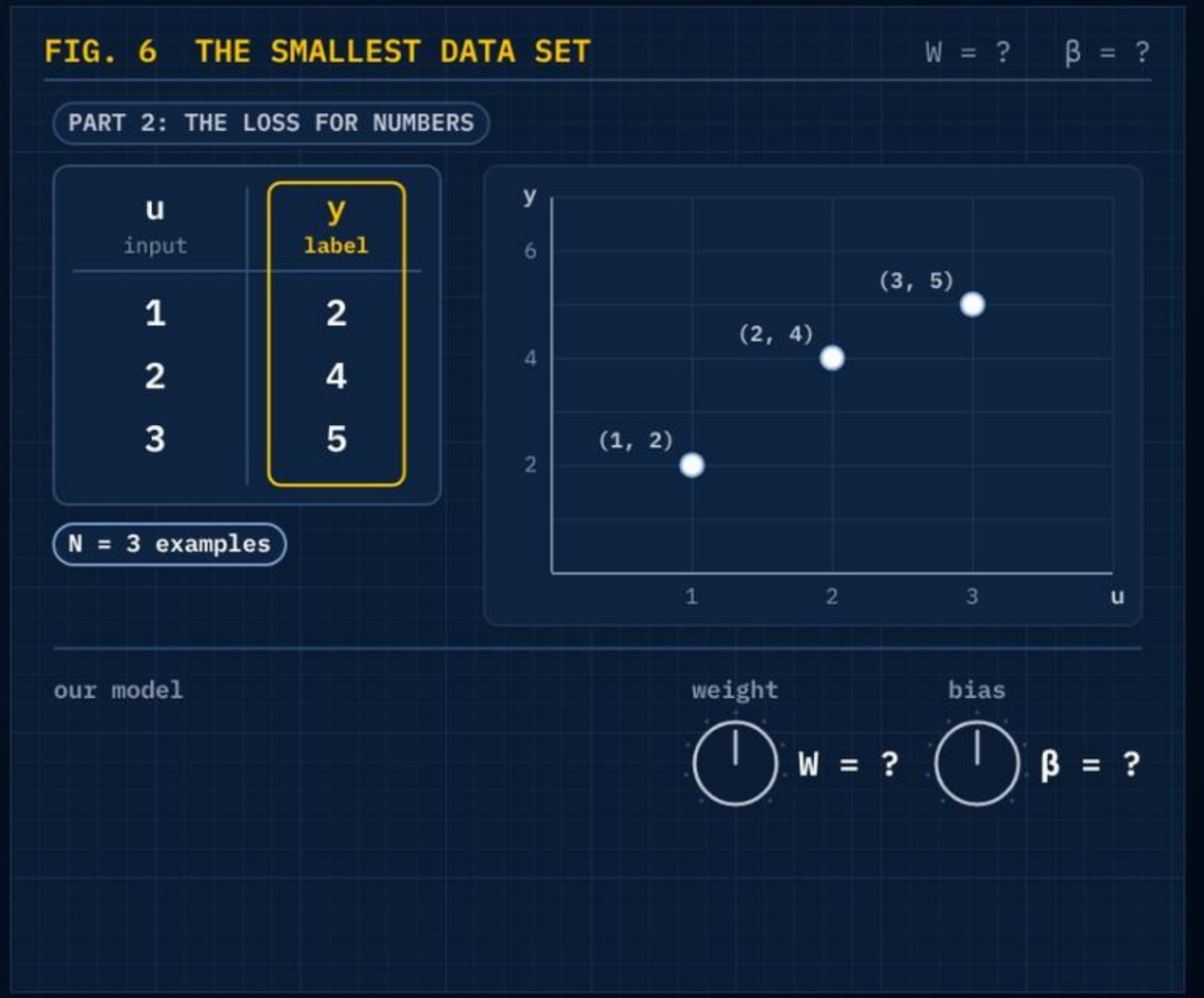
Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft, ch. 7 (§§7.6–7.8). Holding θ fixed at inference is this series’ definition.

Three examples

$(u, y) = (1, 2), (2, 4), (3, 5)$

01 y is the label: the right answer

02 Model: $\hat{y} = W \cdot u + \beta$



The three points are a teaching example. Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND

Line A: W = 2, β = 0

Predict, then subtract: label – prediction.

01 Predictions: 2, 4, 6

02 Errors: 0, 0, –1

FIG. 7 LINE A, BY HAND

ERROR 1 / 3

line A: W = 2, β = 0

predict $\hat{y} = W \cdot u + \beta = 2 \cdot u$

u = 1: 2 × 1 = 2

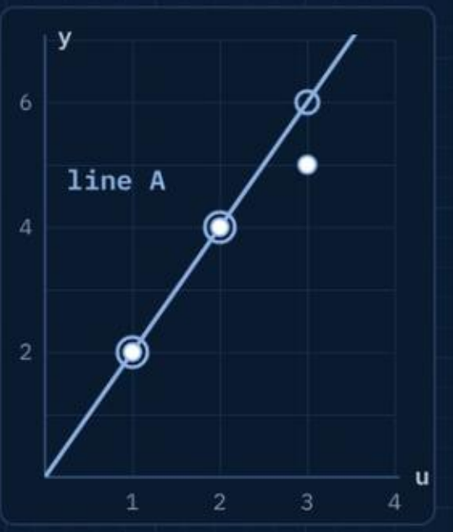
u = 2: 2 × 2 = 4

u = 3: 2 × 3 = 6

errors y – ŷ

label – prediction

u = 1: 2 – 2



● label y (right answer)

○ prediction ŷ (on the line)

The three points are a teaching example. Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND

Square, add, divide

$$L = (1/N) \Sigma (y - \hat{y})^2$$

01 Squares: 0, 0, 1

02 Squaring: misses above and below can't cancel

03 Sum 1 ÷ N = 3 → MSE = 1/3 ≈ 0.333

FIG. 8 SQUARE THE ERRORS

ADD, ÷ N

square each error

u = 1: $0^2 = 0$

u = 2: $0^2 = 0$

u = 3: $(-1)^2 = 1$

add: 0 + 0 + 1 = 1

divide by



why square?



both count

plain: −1 + 1 = 0

squared: 1 + 1 = 2

Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 (maximum likelihood), eq. 5.66 (mean squared error). The three points are a teaching example. Worked example; every number is checked in the companion notebook.

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PART 2 · BY HAND

Line B: W = 1.5, β = 2/3

Oddly specific numbers. Hang on to that.

01 Predictions: 13/6, 11/3, 31/6

02 Errors: −1/6, +1/3, −1/6

FIG. 9 LINE B, BY HAND

ERROR 1 / 3

line B: W = 1.5, β = 2/3

oddly specific: hang on

predict ŷ = 1.5·u + 2/3

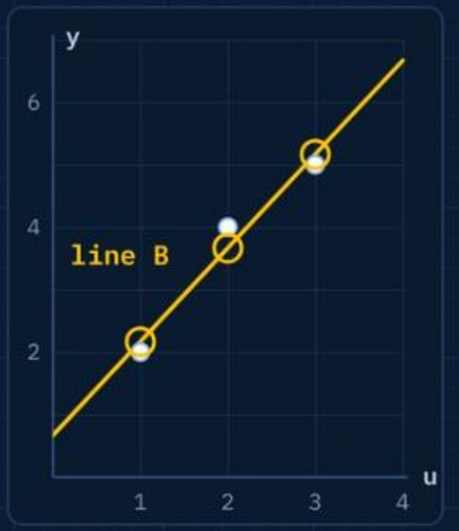
u = 1: 1.5 + 2/3 = 13/6

u = 2: 3 + 2/3 = 11/3

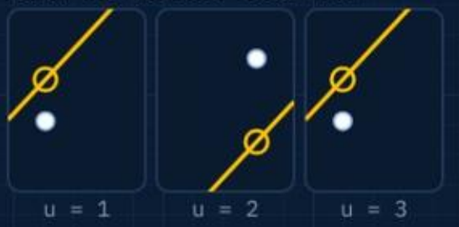
u = 3: 4.5 + 2/3 = 31/6

errors y − ŷ

label − prediction



zoom ×4 around each dot



The three points are a teaching example. Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND

Line B's loss

Square, add, divide by three.

01 Squares: 1/36, 4/36, 1/36

02 Sum 6/36 = 1/6

03 ÷ 3 → MSE = 1/18 ≈ 0.056: six times smaller

FIG. 10 LINE B: SQUARE IT

MSE = 0.056

square each error

errors: −1/6, +1/3, −1/6

u = 1: $(1/6)^2 = 1/36$

u = 2: $(1/3)^2 = 1/9 = 4/36$

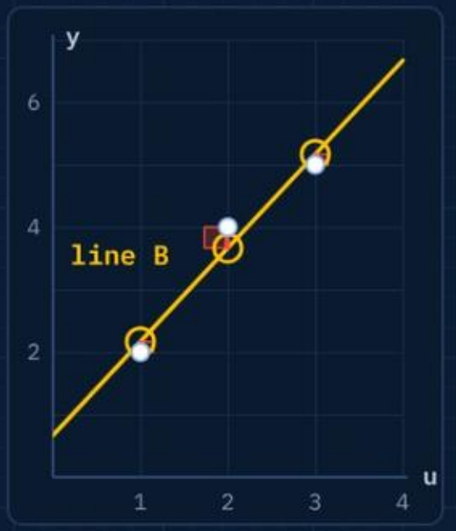
u = 3: $(1/6)^2 = 1/36$

sum: 1/36 + 4/36 + 1/36

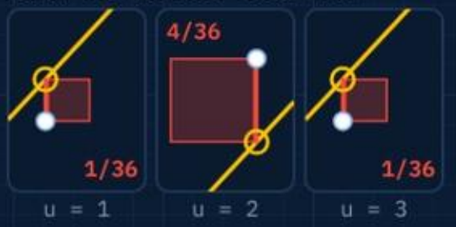
= 6/36 = 1/6

÷ 3: 1/6 ÷ 3 = 1/18

MSE = 1/18 ≈ 0.056



zoom ×4 around each dot



The three points are a teaching example. Worked example; every number is checked in the companion notebook.

PART 2 · THE ANSWER

A little off everywhere wins

Line B is the least-squares fit for these three points.

01 No other W and β beats 1/18

02 Line C (W 1.5, β 1/2): errors 0, 1/2, 0

FIG. 11 A VS B, AND LINE C

LINE C

the answer to the cold open

B: a little off everywhere

beats

A: perfect twice, wrong once

by a full unit

not luck: least-squares fit

no other W, β beats 1/18

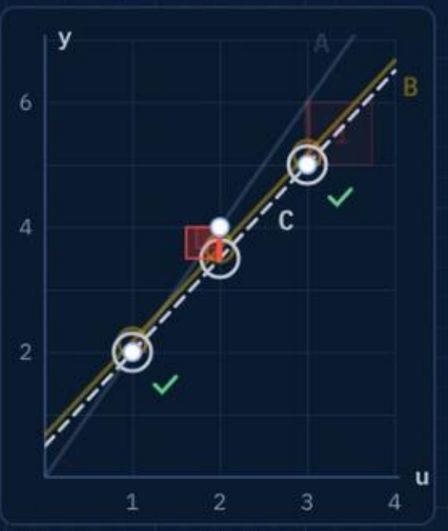
try line C: W = 1.5, β = 1/2

ŷ = 2, 3.5, 5

errors: 0, 1/2, 0

squares: 0, 1/4, 0

1/4 ÷ 3



MSE, lower is better

A 0.333

B 0.056

The three points are a teaching example. Worked example; every number is checked in the companion notebook.

PART 2 · A PICTURE

Think of darts

Distance from the bullseye, squared, averaged over throws.

01 Line A: bullseye twice, then a big miss



Analogy built for this series. The three points are a teaching example. Worked example; every number is checked in the companion notebook.

A model predicts a word

After “the cat sat on the”, it hands out probabilities.

01 So what is a miss?

02 mat 0.30 · floor 0.20 · sofa 0.15



Episode 1’s probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

PART 3 · CROSS-ENTROPY

Only one term survives

$H(q, p) = -\Sigma\, q \ln p$, with $q = 1$ on mat, 0 elsewhere

01 $1 \cdot \ln 0.30 + 0 \cdot \ln 0.20 + 0 + 0$

FIG. 14 ONE-HOT TARGET — TERM 4 / 4

word	observed	mat	floor	sofa	other
p illustrative		0.30	0.20	0.15	0.35
q target		1	0	0	0

$H(q, p) = -\Sigma\, q \ln p$ — Episode 4: cross-entropy

four terms

$$-(1 \cdot \ln 0.30 + 0 \cdot \ln 0.20 + 0 \cdot \ln 0.15 + 0 \cdot \ln 0.35)$$

Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 3 §3.13 (cross-entropy). Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

MARK

PART 3 · BY HAND

−ln 0.30, worked out

$\ln(3/10) = \ln 3 - \ln 10$

01 $\ln 3 \approx 1.099$, $\ln 10 \approx 2.303$

02 $1.099 - 2.303 = -1.204$

03 Surprisal: 1.204 nats (as in Episode 4)



Worked example; every number is checked in the companion notebook. Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

PART 3 · THE LOSS

One term, then the average

$$L = (1/N) \Sigma -\ln p(\text{observed word})$$

01 Next word at 0.60: −ln 0.6 = 0.511

02 1.204 + 0.511 = 1.715

FIG. 16 ONE TERM VS THE LOSS — AVERAGE N = 2

term 1: mat p = 0.30 → 1.204

term 2: next word p = 0.60

ln 0.6 = ln 6 − ln 10

1.792 − 2.303 = −0.511

−ln 0.60 = 0.511 nats

average the two

1.204 + 0.511 = 1.715



Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft, ch. 7 (§§7.6–7.8). Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

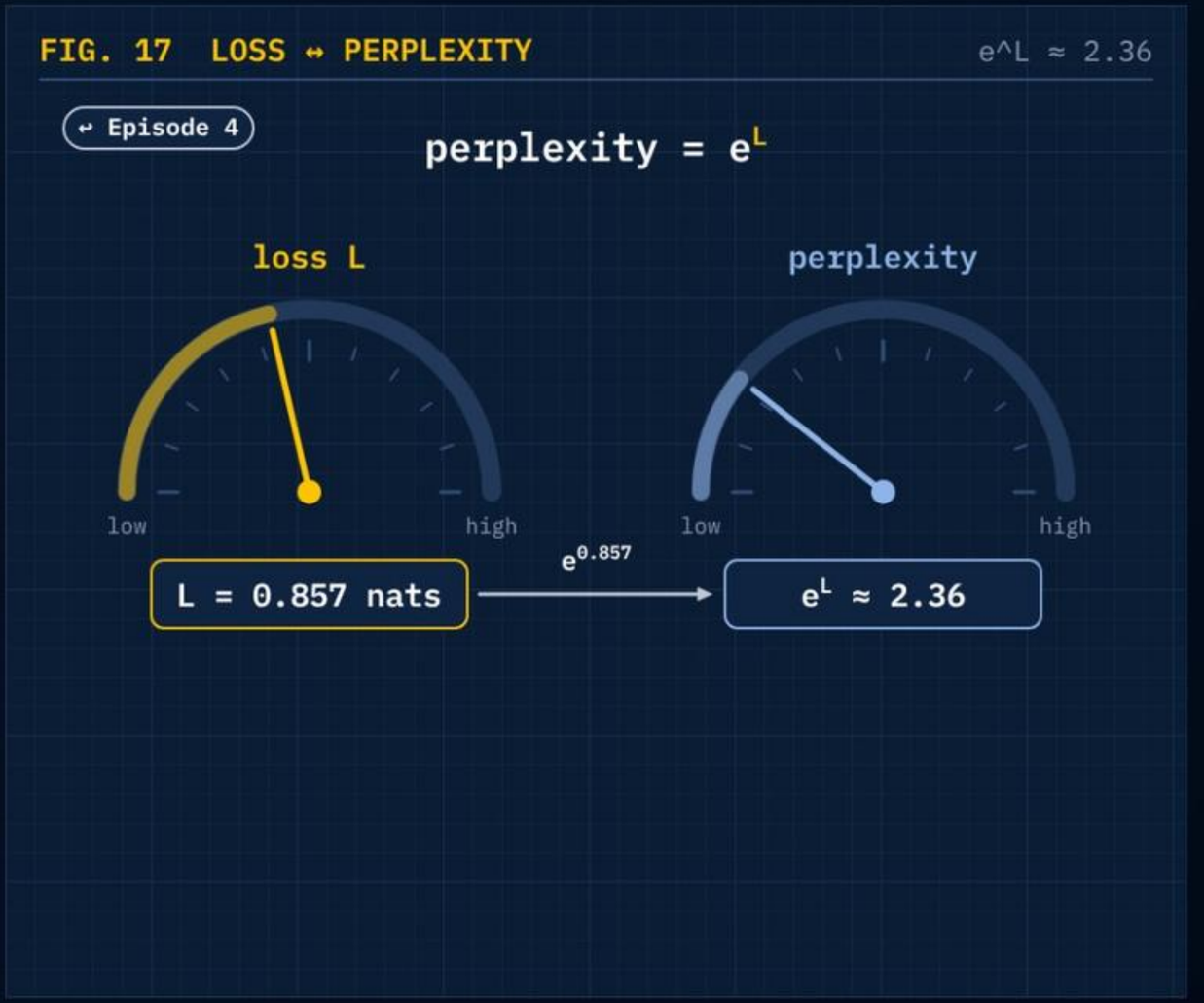
PART 3 · CALLBACK

Loss and perplexity

Perplexity = e to the mean surprisal (Episode 4).

01 e^0.857 ≈ 2.36

02 Same measurement, two units



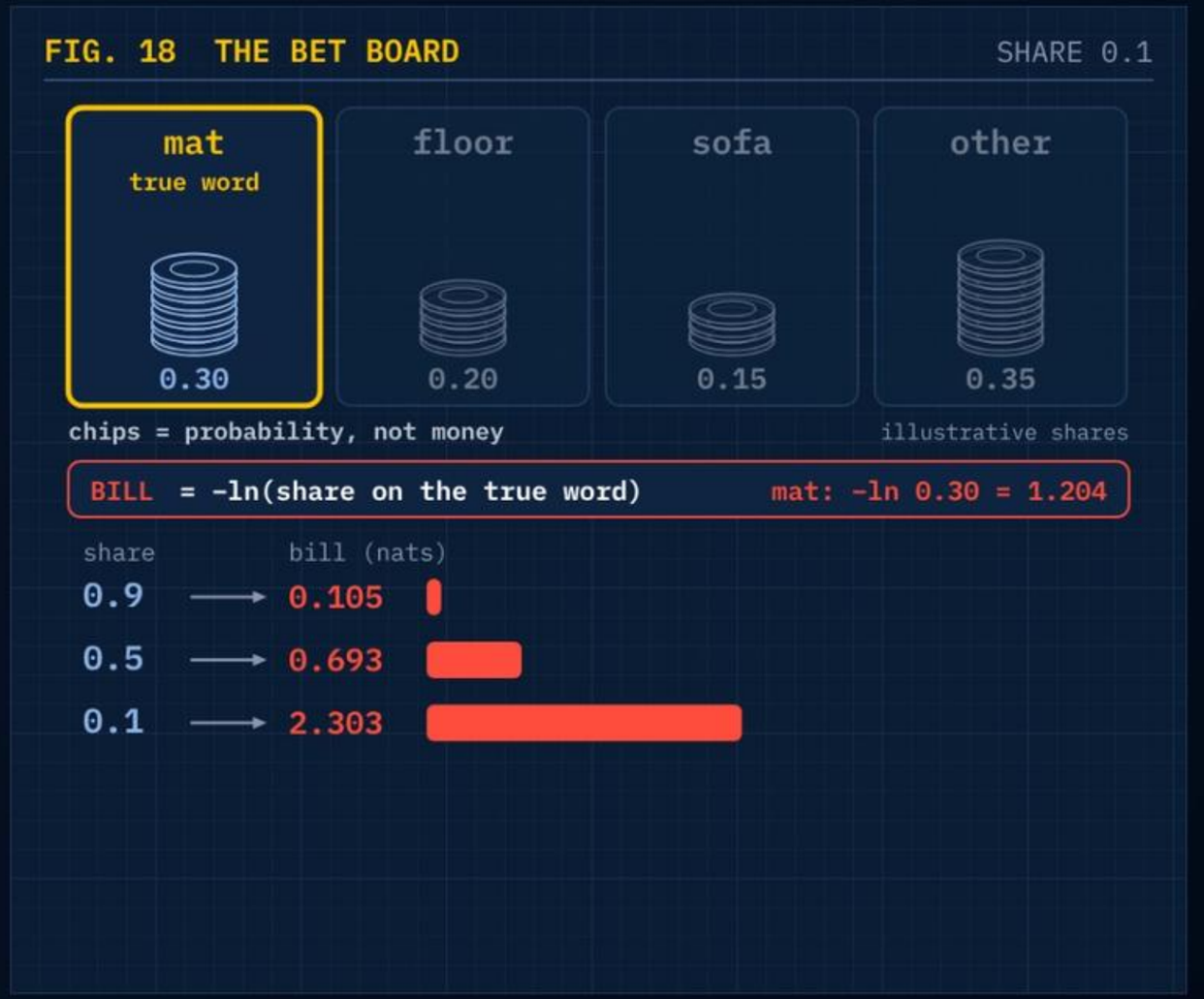
Episode 4 (perplexity). Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft, ch. 7 (§§7.6–7.8).

Think of a bet

Spread your chips; pay −ln of your share on the true word.

01 0.9 → 0.105 · 0.5 → 0.693

02 0.1 → 2.303 · 0.01 → 4.605



Analogy built for this series. Worked example; every number is checked in the companion notebook.

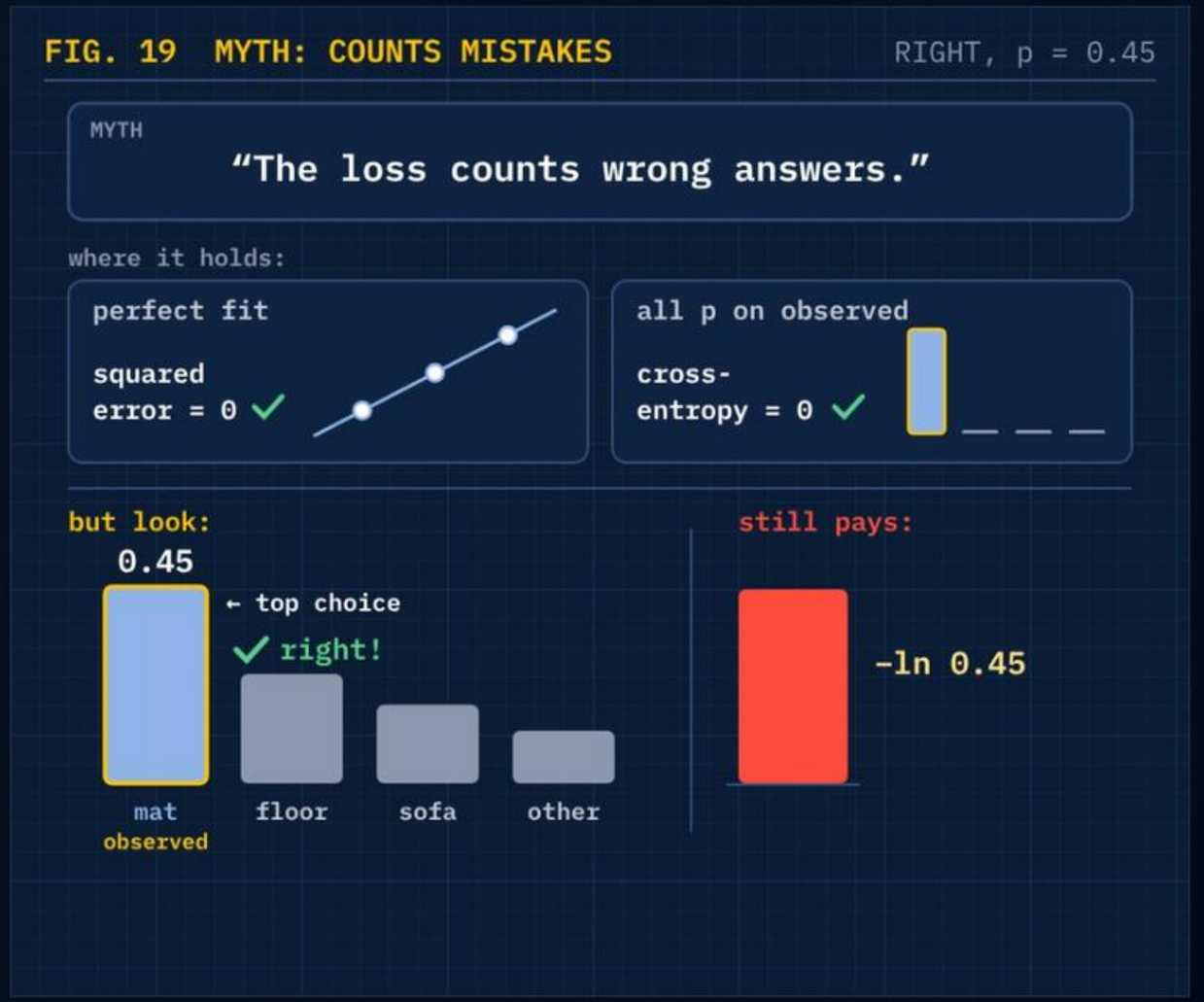
MYTH CHECK

“The loss counts wrong answers”

Something to it: a perfect fit pays zero.

01 Right at 0.45 still pays –ln 0.45 = 0.799

02 Any probability below 1 costs something



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §6.2.1.1 (cross-entropy cost). Worked example; every number is checked in the companion notebook.

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RECAP

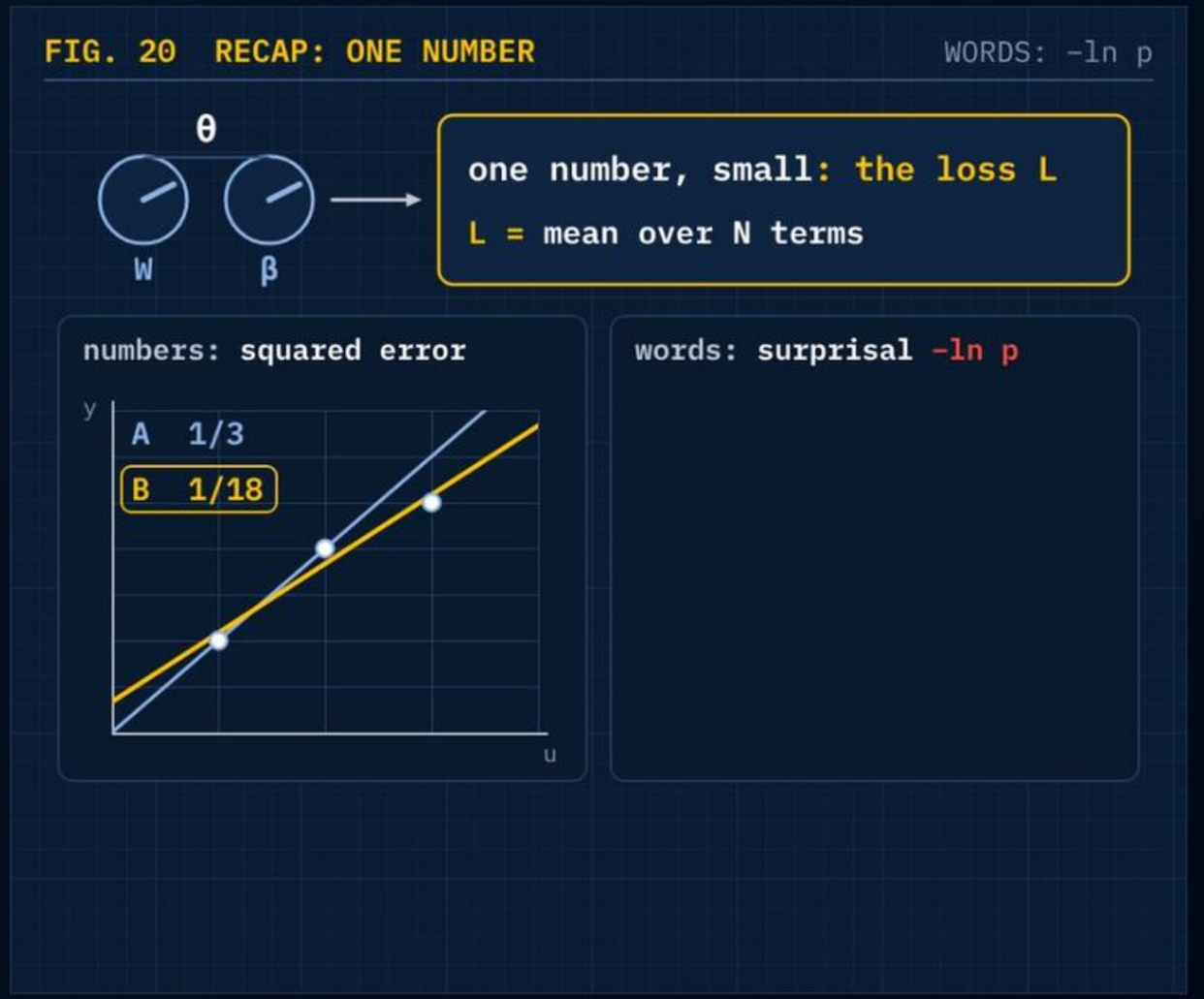
One number to push down

Learning turns the knobs θ to make the loss small.

01 The loss: the mean over N terms

02 Numbers: squared error (1/3 vs 1/18)

03 Words: −ln p (1.204, 0.511 → 0.857)



All sources are listed in the description.

MARK

YOUR TURN

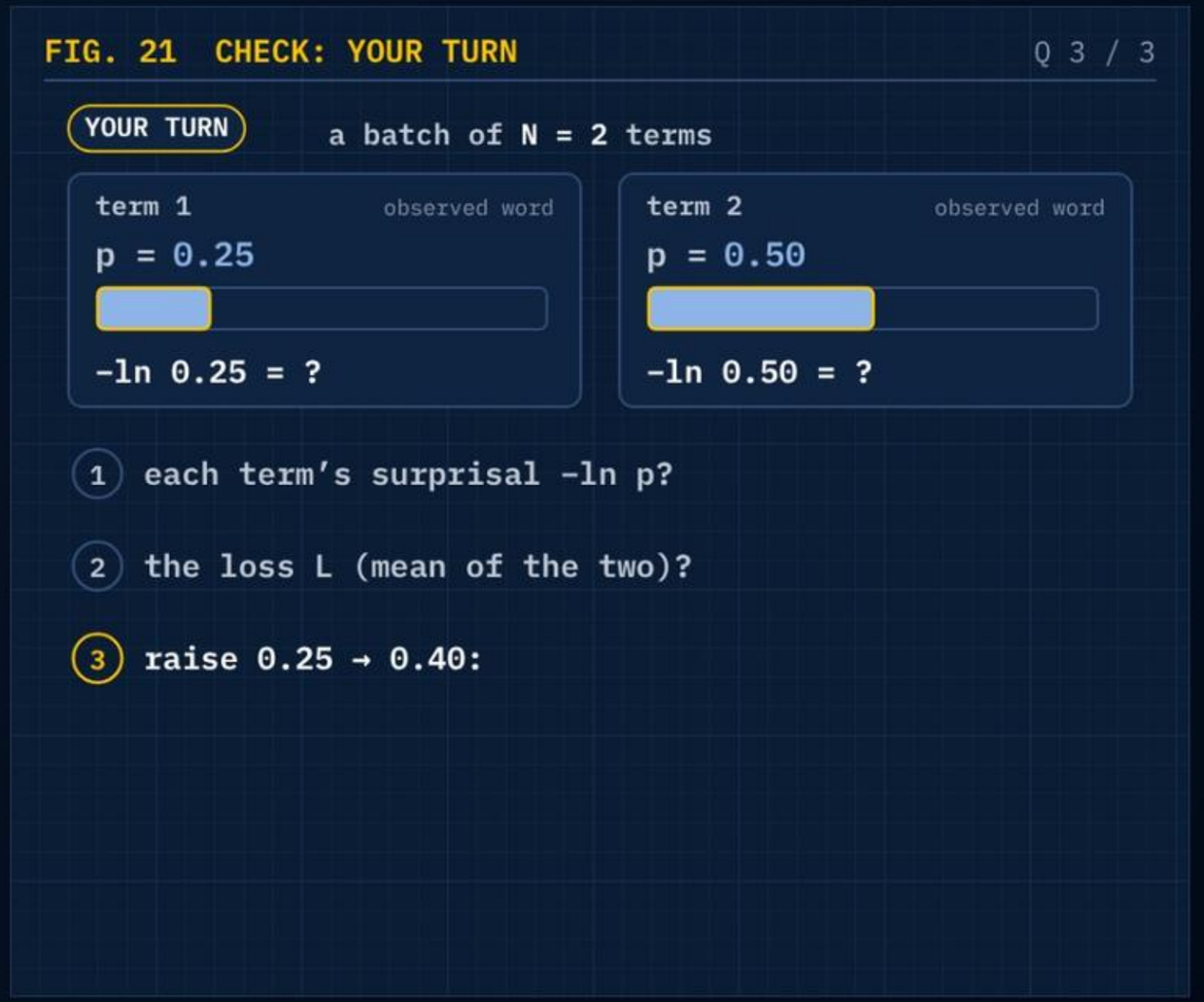
One question to take with you

N = 2. The observed word gets 0.25, then 0.50.

01 Each term’s surprisal?

02 The loss L?

03 Raise 0.25 to 0.40: does L go down?



Worked example; every number is checked in the companion notebook.

YOUR TURN

The answer

One nudge at a time, that's what training does.

01 1.386 and 0.693

02 L = 1.040 nats

03 −ln 0.40 = 0.916 → L = 0.805: yes

FIG. 22 ANSWER: BY HAND

Q3 RAISE p_1

N = 2, p = 0.25 and 0.50

−ln 0.25 = 1.386

−ln 0.50 = 0.693

L = (1.386 + 0.693) / 2 = 1.040 nats

raise the first: 0.25 → 0.40

−ln 0.40 = 0.916

L = (0.916 + 0.693



Worked example; every number is checked in the companion notebook.

TRY THIS WEEK

Three things to try.

01

Pick your own W and β for the three dots. Can you beat 1/18?

02

In the notebook, slide the probability on mat and watch its surprisal

03

Compute the loss for three words, then e to the loss: perplexity

Companion notebook and every source in the description.

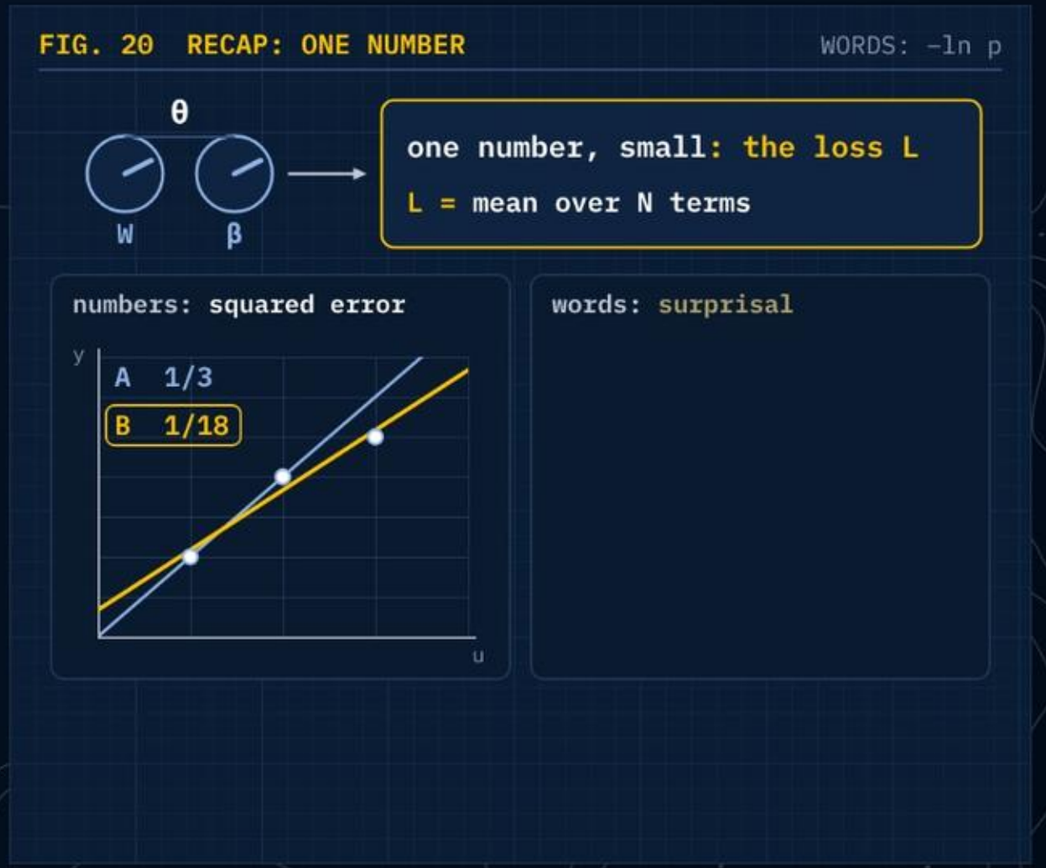
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NEXT TIME

Episode 8: Why these two losses? 0.30 × 0.60, a log, and maximum likelihood.

In the comments: did you expect the line that misses every dot to win?

How Large Language Models Work · subscribe or follow for Episode 8



MARK